Customer Shopping Behavior Analysis

Transforming data into strategic insights for retail optimization

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Project Overview

Objective

Uncover purchasing patterns and behavioral trends from 3,900 transactions

Key Question

How can consumer data identify trends and optimize marketing strategies?

Tools

Python, SQL, and Power BI for complete data pipeline

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Dataset Summary

3,900 Purchases

18 columns of demographic and behavioral data

- Customer demographics: Age, Gender, Location, Subscription
- Purchase details: Product, Category, Price, Season, Size
- Behavioral data: Discounts, Promo codes, Ratings, Shipping

Data Quality

37 missing values in Review Rating handled during preprocessing

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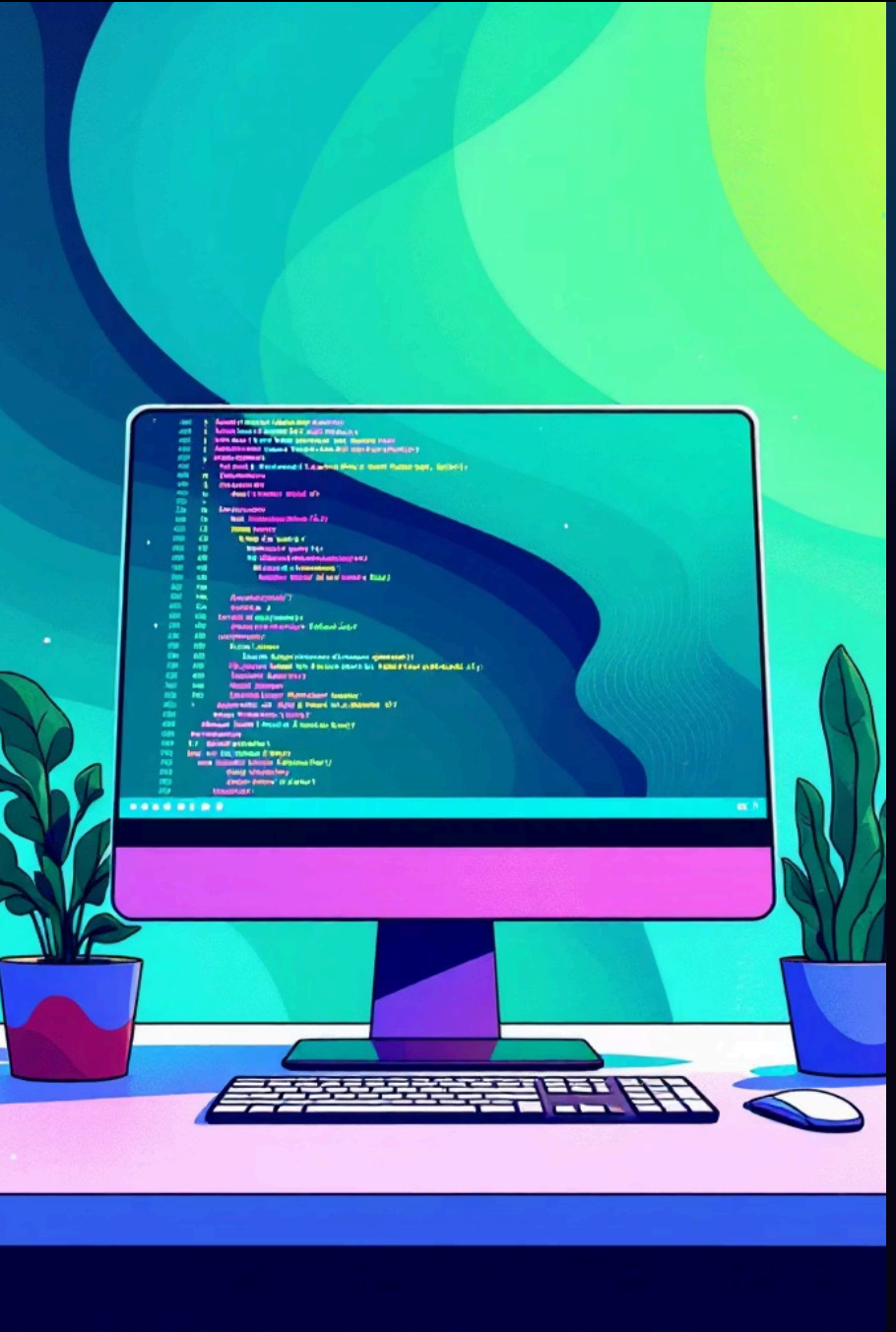

Data Preparation Process

Data Loading

Imported using pandas

Exploratory Analysis

info() and describe() methods

Data Cleaning

Handled missing values, standardized formats

Database Load

PostgreSQL integration

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SQL Analysis Highlights

1

Revenue by Gender

Male: $157,890 | Female: $75,191

2

High-Spending Discount Users

839 customers spending above average with discounts

3

Top Products by Rating

Gloves (3.86), Sandals (3.84), Boots (3.82)

4

Shipping Analysis

Express: $60.48 | Standard: $58.46

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Key SQL Findings

Subscribers vs Non-Subscribers

Non-subscribers: 2,847 customers, $170,436 revenue

Subscribers: 1,053 customers, $62,645 revenue

Discount-Driven Products

Hat (50%), Sneakers (49.66%), Coat (49.07%)

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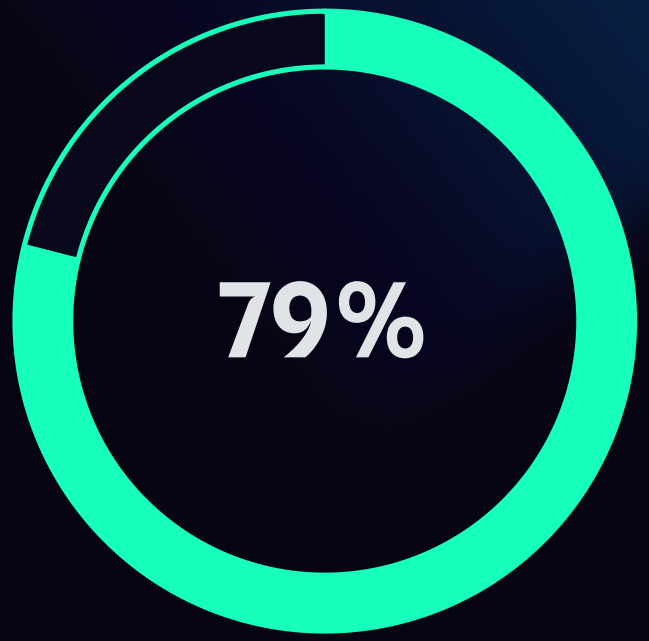
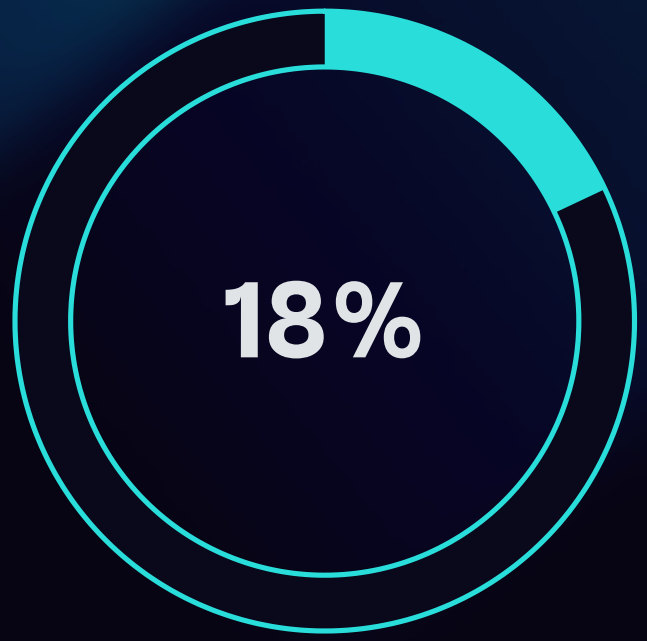
Customer Segmentation



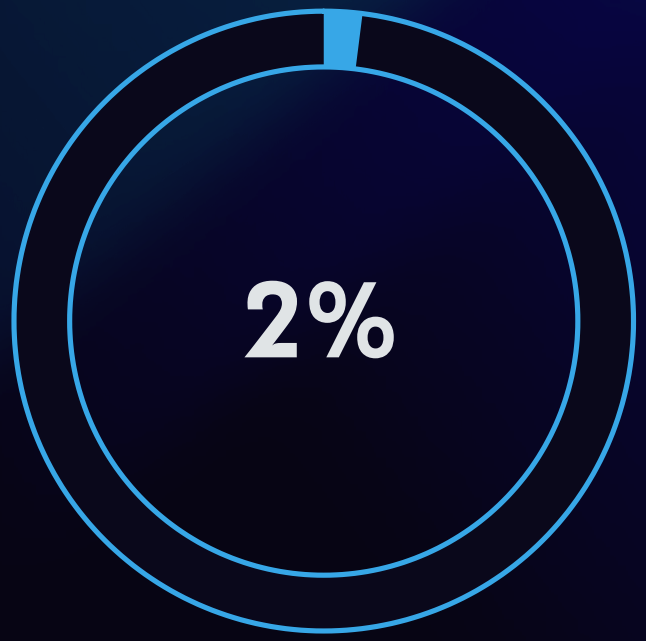
Loyal Customers

3,116 customers with repeat purchases



Returning

701 customers making second purchases



New

83 first-time customers

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Power BI Dashboard

Interactive Filters

Age group, Category, Gender, Subscription status

KPI Cards

Revenue and customer metrics at a glance

Visual Comparisons

Bar charts, pie charts, line graphs for trends

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Business Recommendations

Enhance Subscriptions

Exclusive discounts and benefits to increase sign-ups

Strengthen Loyalty

Reward systems for repeat customers

Optimize Discounts

Balance usage to maximize revenue

Promote Top Products

Focus on high-rated, best-selling items

Targeted Campaigns

Personalize by age groups, frequent buyers, express users

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Conclusion

Combining **Python, SQL, and Power BI** transforms raw data into actionable business insights

Understand Behavior

Comprehensive customer analysis

Improve Engagement

Data-driven customer strategies

Drive Growth

Optimized marketing and long-term success

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